SeedheBol

Functional Indian languages, offline, for people who move for work.

On-device, voice-first language learning for India's 140 million internal migrant workers — built for the phone they already own, the language they already speak, and the life they're already living.

THE PERSON WE ARE BUILDING FOR

Meet Ramesh.

34

Ramesh

Construction worker

FROM	Muzaffarpur, Bihar
SPEAKS	Bhojpuri & functional Hindi
READS	Devanagari only, haltingly
WORKING IN	Chennai — six-month contract
DEADLINE	Three weeks to speak functional Tamil
OR ELSE	Supervisor stops assigning skilled work

"He can't read Tamil. He can barely read Hindi."

This is the constraint every existing language app ignores — and the one that shapes this entire product.

- Site safety vocabulary
- Wage negotiation
- Buying food & directions
- Hospital basics

THE SCALE OF IT

Ramesh is not alone.

140M	450M	30%	0
internal migrant workers today	internal migrants — 37% of India	of migrant women cannot read	consumer products built for them

Bihar → Tamil Nadu. Odisha → Kerala. UP → Karnataka. Every major migration corridor in India crosses a language boundary — with essentially zero learning material and no shared script.

WHY THIS MATTERS

Not speaking the local language isn't an inconvenience.

01

Wage suppression

Can't negotiate, verify piece rates, or challenge underpayment.

02

Safety risk

Can't read site safety instructions or describe an injury at a hospital.

03

Exploitation exposure

Total dependence on a contractor as the sole interface to the outside world.

04

Service exclusion

Can't navigate a ration shop, a bank, a police station, a school admission.

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It is also a mental-health cost nobody measures: social isolation, in a place you cannot yet call home.

THE COMPETITIVE GAP

Why every existing option fails him.

Duolingo	English-centric. Assumes literacy. Teaches tourist words, not survival vocabulary
Google Translate	A lookup tool, not a teacher. No retention, no pronunciation feedback.
Bhashini-powered apps	Excellent models — but cloud-first. No offline mode at all.
NGO / in-person classes	Actually work — but don't scale, and shift work makes fixed timing impossible.

The unoccupied position: Indian-language-to-Indian-language, offline, voice-first, occupation-specific — on the phone he already owns.

THE REFRAME

Duolingo teaches you to talk about owls.
India's language problem is a Bihari construction worker in Chennai who has three weeks to become functional in Tamil.

Everything in SeedheBol follows from taking that user seriously.

PRODUCT THESIS

Three non-negotiable design principles.

Voice is the interface, not a feature

Every core loop is completable without reading a single word. Text is an optional enhancement layer.

Direct L1 → L2, never through English

A Bhojpuri speaker learning Tamil gets explanations in Bhojpuri and Tamil. English pivoting destroys register and idiom.

Offline is the default state

Not a degraded mode — the assumption. Connectivity, when present, is used only for optional sync, never for inference.

THE MOAT

On-device is not an optimisation. It's what makes the flagship feature legal.

Ambient vocabulary mining

The phone passively hears the language around Ramesh — a shop, a canteen, a bus — and builds tomorrow's lesson from it.

Streaming continuous ambient audio to a cloud server is a privacy catastrophe and a non-starter under India's DPDP framework.

On-device is what makes this feature shippable at all.

Zero marginal cost

Free forever — no per-query API bill

Zero data cost

Works on a rationed 1.5GB/day plan

Zero latency floor

Roleplay feels like conversation

Works in airplane mode

Construction site, no signal, no problem

WHAT WE BUILT

Six features. Nothing else, until these ship.

01

Offline roleplay dialogue loop

The demo. If this works, nothing else is essential.

02

Phoneme-level pronunciation feedback

L1-interference-aware targeting — the technical depth.

03

Camera OCR → micro-lesson

Point at a signboard, get a translation and a lesson.

04

Ambient vocabulary mining

The moat argument — impossible as a cloud product.

05

Zero-literacy voice-first shell

The entire app operable without reading anything.

06

Office Kit teacher export

Batch assessment report, phone to laptop.

UNDER THE HOOD

Two MIT-licensed models. ~100MB. Nothing leaves the phone.

IndicConformer

ASR + forced alignment

~120M params → ~60MB INT8

Hybrid CTC-RNNT: one model gives both the transcript and the posteriorgram pronunciation scoring needs.

FastPitch + HiFi-GAN V1

On-device runtime TTS

~25–40M params → ~40MB

Chosen for intelligibility over naturalness — AI4Bharat's own benchmark shows lower CER than VITS.

No backend server

No database

No cloud API calls during inference

No analytics SDK that phones home

Runs on Qualcomm Hexagon NPU (Snapdragon 8 Elite Gen 5, iQOO 15) via ONNX Runtime + QNN.

THREE MINUTES. ONE NARRATIVE.

The demo.

0:00 Meet Ramesh — 34, three weeks to speak enough Tamil to keep his job.

0:15 Airplane mode, on stage. Nothing here touches the internet.

0:45 Live roleplay with a Tamil site supervisor — interrupt it mid-sentence, it listens.

1:20 Pronunciation feedback: "your ட came out as a த" — named, because it knows his L1 is Bhojpuri.

1:50 Point the camera at a Tamil signboard — instant translation, instant five-word lesson.

2:15 Ambient mining: the words this room just spoke, already queued for tomorrow's lesson.

3:00 Close: "Still in airplane mode."

BEYOND THE HACKATHON

140 million people are waiting for software built for them.

- Free for individuals, always — zero marginal inference cost makes it sustainable
- Employer & NGO distribution — construction firms, hospitals, migrant welfare unions
- Verified skill attestation — a learning app that becomes an employability credential

Zero literacy required. Zero connectivity required. Zero excuse left.

SeedheBol — built for Ramesh.

Thank you & QnA